

Comprehensive system/service non-human user report with disk quota, usage, and activity

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Generates a comprehensive system audit report for all non-human (system/service) users (requires sudo) showing username, disk quota limit, home directory path, actual disk usage, account creation date, file count, last login timestamp, and last executed command. All information is formatted in a readable table with elevated privileges for complete data access.

Language

bash

Category

system-administration

Command

```
sudo bash -c "printf 'USER\tQUOTA\tHOME\tDISK_USAGE\tCREATED\n\tFILES\tLAST_LOGIN\tLAST_CMD\n'; getent passwd | awk -F: '
  \$3 < 1000 || \$1 ~ /^nobody$/ { print \$1,\$6 }' |
  while read u h; do q=\$(quota -u \"\$u\" 2>/dev/null |
  tail -1 | awk '{print \$2}'); [ -z \"\$q\" ] && q=none; d=
  \$(du -sh \"\$h\" 2>/dev/null | cut -f1); c=\$(stat -c %
  y \"\$h\" 2>/dev/null | cut -d' ' -f1 || echo N/A); f=\$(
  find \"\$h\" -type f 2>/dev/null | wc -l); l=\$(lastlog -
  u \"\$u\" 2>/dev/null | tail -1 | awk '{print \$5,\$6,\$7
  }'); [ -z \"\$l\" ] && l=Never; cmd=\$(tail -1 \"\$h\"/.
  bash_history 2>/dev/null | head -c 30); [ -z \"\$cmd\" ]
  && cmd=N/A; printf '%s\t%s\t%s\t%s\t%s\t%s\t%s\t%s\n' \"\$
  u\" \"\$q\" \"\$h\" \"\$d\" \"\$c\" \"\$f\" \"\$l\" \"\$
  cmd\"; done" | column -t -s $'\t'
```

Explanation

The command uses 'sudo bash -c' to execute the entire script with elevated privileges, ensuring access to all user data. It retrieves users with UID ≤ 1000 (or 'nobody') via getent, then for each user gathers: disk quota from 'quota -u', home disk usage from 'du -sh', account creation date via 'stat', file count using 'find', last login from 'lastlog', and last command from .bash_history. The 'bash -c' wrapper allows proper variable expansion and quoting within the sudo

context. Output is joined by tabs and formatted by 'column -t' for readability, avoiding issues with spaces in values.

Tags

user-management, disk-usage, system-administration, activity, report, quota, audit, system-users, service-accounts

Dependencies

coreutils, util-linux, shadow-utils, findutils

Arguments

1. **UID_THRESHOLD** (Optional): Maximum UID to consider a system user (default: 1000 for non-human users)
Default: 1000

Examples

1.

```
sudo bash -c "printf 'USER\tQUOTA\tHOME\tDISK_USAGE\tCREATED\tFILES\tLAST_LOGIN\t' >/dev/null | tail -1 | awk '{print \$2}'; [ -z \"\$q\" ] && q=none; d=\"\$(du -sh \"\$h\" 2>/dev/null | cut -f1); c=\"\$(stat -c %y \"\$h\" 2>/dev/null | cut -d' ' -f1 | wc -l); f=\"\$(find \"\$h\" -type f 2>/dev/null | wc -l); l=\"\$(lastlog -u \"\$u\" 2>/dev/null | tail -1 | awk '{print \$5, \$6, \$7}'); [ -z \"\$l\" ] && l=Never; cmd=\"\$(tail -1 \"\$h\"/.bash_history 2>/dev/null | head -c 30); [ -z \"\$cmd\" ] && cmd=N/A; printf '%s\t%s\t%s\t%s\t%s\t%s\t%s\t%s\t' \"\$u\" \"\$q\" \"\$h\" \"\$d\" \"\$c\" \"\$f\" \"\$l\" \"\$cmd\"; done" | column -t -s $'\t' - Generate complete report for all non-human (system) users with all metrics (requires sudo)
```
2.

```
sudo bash -c "printf 'USER\tQUOTA\tDISK_USAGE\tCREATED\tFILES\tLAST_LOGIN\t' >/dev/null | tail -1 | awk '{print \$2}'; [ -z \"\$q\" ] && q=none; d=\"\$(du -sh \"\$h\" 2>/dev/null | cut -f1); c=\"\$(stat -c %y \"\$h\" 2>/dev/null | cut -d' ' -f1 | wc -l); f=\"\$(find \"\$h\" -type f 2>/dev/null | wc -l); l=\"\$(lastlog -u \"\$u\" 2>/dev/null | tail -1 | awk '{print
```

```
\{$5,\{$6,\{$7\}''); [ -z \{"\{$1\}" ] && l=Never; printf '%s\t%s\t%s\t%s\t%
\{"\{$u\}" \{"\{$q\}" \{"\{$d\}" \{"\{$c\}" \{"\{$f\}"
\{"\{$l\}"; done | sort -t $'\t' -k3 -hr" | column -t -s $'\t'
- Sort by disk usage descending, exclude home path and last command
```

3. `sudo bash -c "printf 'USER\tQUOTA\tDISK_USAGE\tFILES\n';
getent passwd | awk -F: ' \{$3 < 1000 || \{$1 ~ /\^nobody$/ { print
\{$1,\{$6 \}' | while read u h; do q=\{$(quota -u \{"\{$u\}"
2>/dev/null | tail -1 | awk '{print \{$2\}'}; [-z \{"\{$q\}"
] && q=none; d=\{$(du -sh \{"\{$h\}" 2>/dev/null | cut -f1);
f=\{$(find \{"\{$h\}" -type f 2>/dev/null | wc -l); printf
'%s\t%s\t%s\t%s\n' \{"\{$u\}" \{"\{$q\}" \{"\{$d\}"
\{"\{$f\}"; done" | column -t -s $'\t' - Simplified report show-
ing only quota, disk usage, and file count`

Output

USER	QUOTA	HOME	DISK_USAGE	CREATED
FILES	LAST_LOGIN	LAST_CMD		
root	none	/root	1.5G	2023-01-15
15340	May 20 13:45	apt update		
daemon	none	/usr/sbin	32K	2023-01-15 12
	Never	N/A		
bin	none	/bin	0	2023-01-15 0
	Never	N/A		
www-data	none	/var/www	4.2G	2023-05-10
45678	Never	N/A		
nobody	none	/nonexistent	0	2023-01-15 0
	Never	N/A		

Notes

- This command is designed to run with sudo for complete information; running without sudo will show incomplete results
- The 'sudo bash -c' wrapper preserves the entire command context and allows proper variable expansion
- Disk quota requires quota system to be enabled on the filesystem; shows 'none' if unavailable
- File count calculation uses 'find' which can be slow on directories with millions of files
- Account creation date uses home directory birth time (%w) if available, falling back to modification time (%y)

- Last command is truncated to 30 characters; extracted from `.bash_history`.
Note: Bash only flushes history to disk upon session exit or when `'history -a'` is run, so this shows the last saved command, not necessarily the active session's most recent command.
- Users with UIDs $\neq 1000$ are standard human accounts and are excluded by default. Note that system/service accounts typically lack valid login shells or history files, making `LAST_CMD` frequently N/A and `LAST_LOGIN` Never.
- The command uses `getent` instead of `/etc/passwd` for better compatibility with different user databases (LDAP, NIS, etc.)

Warnings

- Requires sudo access to function properly; user must have sudo privileges without password prompt configured in `sudoers` for unattended execution
- Large file counts or extensive directory trees will cause significant performance degradation
- Running with `find` on large home directories can consume considerable IO and CPU resources
- The `lslogins` command retrieves last login information depending on `wtmp`/`btmptmp` logs; accuracy varies by distribution logging policies
- Last command information depends on `.bash_history` being available and enabled; other shells use different history files
- Network-mounted home directories (NFS, SMB) will significantly increase command execution time
- On systems with many users, overall execution time can be substantial; consider running during off-peak hours

See Also

- `find-largest-storage-users`
- `du-sort-largest-directories`
- `disk-space-usage-per-directory`
- `find-large-files-recursive`

Status

reviewed

Safety

caution

Shell

bash

Platforms

linux, gnu-linux

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